Rahat Rizvi Rahman

Richmond, VA, USA
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Portfolio: https://rahatrizvirahman.github.io/

Research Interests

Natural Language Processing, Machine Learning, Mining Software Repositories, Empirical Software Engineering, Human–AI Interaction, Multi-turn Conversations, and Failure Analysis in AI-assisted Development.

Education

Virginia Commonwealth University (VCU) Richmond, VA, USA
Ph.D. Student in Computer Science
Research Focus: Natural Language Processing, Machine Learning, Mining Software Repositories, Empirical Software Engineering Aug 2024 – Present

University of Dhaka Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; CGPA: 3.51/4.00 Jan 2017 – Aug 2021
Undergrad Thesis: mMultiSURF - A relief-based feature selection method considering class overlapping area among neighboring instances and prior information.

Research Experience

Research Assistant, Dept. of Computer Science, Virginia Commonwealth University Spring 2025 – Present
Advisor: Prof. Kostadin Damevski
- Conducting research on multi-turn human–AI conversations, focusing on identifying failure points and evaluating mitigation strategies.
- Building NLP pipelines for analyzing large-scale conversation transcripts.
- Exploring how large language models can be applied to empirical studies in software engineering.
- Conducting research on multimodal VR emotion analysis using facial Action Units and audio features collected in controlled lab settings.

Teaching Assistant, Dept. of Computer Science, Virginia Commonwealth University Fall 2024
Course: CMSC-426-001 — Software as a Service
Instructor: Prof. Kostadin Damevski
- Assisted with lectures, assignments, and grading.
- Provided mentorship and guidance to undergraduate students.

Teaching Assistant, Dept. of Computer Science, Virginia Commonwealth University Spring 2026
Course: CMSC-425/525 — Intro Software Analysis & Test
Instructor: Prof. Kostadin Damevski
- Assisting with lectures, assignments, grading, and student mentoring.

Research Work

- **Rahman, R. R.**, Ali, A. A., Rahman, A. K. M. M. *"Bangla Char-BERT: A Hybrid Character and Token-Level Language Model for Low-Resource Bangla"* Manuscript in preparation (2025).
- **Rahman, R. R.** *"mMultiSURF: A Relief-Based Feature Selection Method for Overlapping Class Boundaries"* Undergraduate Thesis, University of Dhaka (2021).

Publications

1. **Rahat Rizvi Rahman**, Mia Mohammad Imran, Kostadin Damevski. *A Low-Cost Human-in-the-Loop Investigation of Toxicity on GitHub at Scale.* In Proceedings of the **41st IEEE/ACM International Conference on Automated Software Engineering (ASE)**, 2026. https://doi.org/10.1145/3832783.3837505
2. Mia Mohammad Imran, Robert Zita, **Rahat Rizvi Rahman**, Preetha Chatterjee, Kostadin Damevski. *Toxicity Ahead: Forecasting Conversational Derailment on GitHub.* In Proceedings of the **IEEE/ACM International Conference on Software Engineering (ICSE)**, 2026.
3. **Rahat Rizvi Rahman**, Hee Yun Choi, Joonghyo Lim, Go Eun Lee, Seungmoo Lee, Chungyean Cho, Kostadin Damevski. *Toward Generalized Emotion Recognition in VR by Bridging Natural and Acted Facial Expressions.* **Sensors** (MDPI), 2026, *26*, 845. https://doi.org/10.3390/s26030845
4. **Rahat Rizvi Rahman**, Seung Pyo Kang, Se Jin Jang, Seungmoo Lee, Chungyean Cho, Kostadin Damevski. *When Pair Programming Gets Emotional: A Case Study of What LLMs Still Miss.* **International Journal of Software Engineering and Knowledge Engineering (IJSEKE)**, World Scientific, 2026. https://doi.org/10.1142/S0218194026500294
5. Terens Tare, **Rahat Rizvi Rahman**, Hee Yun Choi, Joonghyo Lim, Go Eun Lee, Seungmoo Lee, Chungyean Cho, Kostadin Damevski. *Detecting Natural Emotions in Virtual Reality Through Facial Movement Analysis.* In Proceedings of the **31st ACM Symposium on Virtual Reality Software and Technology (VRST)**, 2025. https://doi.org/10.1145/3756884.3766013
(First empirical study comparing natural vs. acted emotions in VR using Meta Quest Pro headset.)

6. Faria Sultana, Md Tahmid Hasan Fuad, Md Fahim, **Rahat Rizvi Rahman**, Meheraj Hossain, M Ashraful Amin, A K M Rahman, Amin Ahsan Ali. *Bangla Text Summarization with Large Language Models: Exploring Abstractive Headline Generation.* In Proceedings of the **International Conference on Pattern Recognition (ICPR)**, pp. 72–86. Springer, Cham, 2025.

Kaggle Competitions

Google Brain - Ventilator Pressure Prediction Sept 2021 – Nov 2021
- Built Conv1D/LSTM/GRU regression models to predict ventilator pressure.
- Ranked 314th of 2605 teams (Top 13%).

Global Wheat Detection May 2020 – Aug 2020
- Applied EfficientDet for wheat head object detection.
- Ranked 323rd of 2245 teams (Top 15%).

Awards and Honors

- 5th place, *Bhashabhrom 2023* Bangla Grammatical Error Detection competition.
- Finalist, Datathon 2.0 organized by Robi (Team: The Anomalies).
- Govt. Scholarship: full tuition waiver and yearly stipend for outstanding high school performance.

Professional Experience

Senior Software Engineer, Enosis Solutions, Dhaka Nov 2023 – July 2024
Software Engineer, Enosis Solutions, Dhaka Nov 2021 – Oct 2023
- Built React-based frontend for Node.js backend apps.
- Configured Microsoft Azure storage and implemented SAS API.
- Designed CI/CD pipelines with GitHub Actions.
- Worked with LangChain & ChatGPT for QA pipelines.
- Developed PDF-to-JSON extraction pipeline using Azure Form Recognizer and ChatGPT.

Selected Projects

Amar Health E-health Tracker (2021) – Node.js-based health record platform.
Purbo Poshcim Online News Portal (2021) – Online news portal with AWS S3 + MongoDB.
ML Algo Simulator (2019) – Flask web-app to visualize gradient descent and ML models.
Share and Learn (2018) – Android app for article sharing with Firebase backend.

Courseworks and Problem Solving

Ph.D. Coursework (VCU): Advanced Natural Language Processing, Machine Learning, Neural Networks and Deep Learning, Software Engineering for AI-Enabled Systems.
Undergraduate Coursework (University of Dhaka): Linear Algebra, Probability and Statistics, Data Mining, Artificial Intelligence, Natural Language Processing.
Problem Solving: Solved 500+ problems across Codeforces, CodeChef, LightOJ, and UVA.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, Latex
Deep Learning: TensorFlow, Keras, PyTorch
Frontend: HTML, CSS, SCSS, React, Redux, jQuery
Backend: Node.js
Databases: MongoDB, Firebase, SQL (Oracle)
Tools: Git, VS Code, IntelliJ, Jupyter Notebook